

COSTVIEW 360

Commercial Construction Intelligence & Field Cost Governance

Authoritative operational guide for Quantity Surveyors, Project Managers, Site Engineers & Financial Controllers.
Complete End-User Handbook · Practical Field Workflows · Visual Process Data Flows · NGN-Native Controls

1. PLATFORM OBJECTIVE: ELIMINATING 15% - 30% COST LEAKAGE

Commercial construction projects in Nigeria routinely suffer from 15% to 30% unbudgeted cost inflation, uncontrolled contractor variation claims, material diversion from site storage, and unverified contractor invoices. CostView 360 bridges the critical operational chasm between physical site operations and commercial finance controllers. By enforcing real-time Bill of Quantities (BOQ) baselines, mandatory 3-way invoice matching, statutory 10% retention withholding, and shift attendance logs, CostView 360 ensures every kobo spent is verified against physical site reality.

TENET 1: NGN (₦) NATIVE VALUATION

Purpose-built for Nigerian construction finance. All BOQ rates, certified sums, and retention funds are natively managed in Naira (₦) with instant dynamic conversions to USD, GBP (£), and EUR (€) for imported MEP equipment and plant.

TENET 2: AUTOMATED 3-WAY INVOICE MATCH

Rigorous cross-verification matching the approved Purchase Order (PO) against physical Goods Received Notes (GRN) from site and the vendor's submitted Invoice. System automatically halts payment on price or quantity discrepancies.

TENET 3: 10% STATUTORY RETENTION FUND

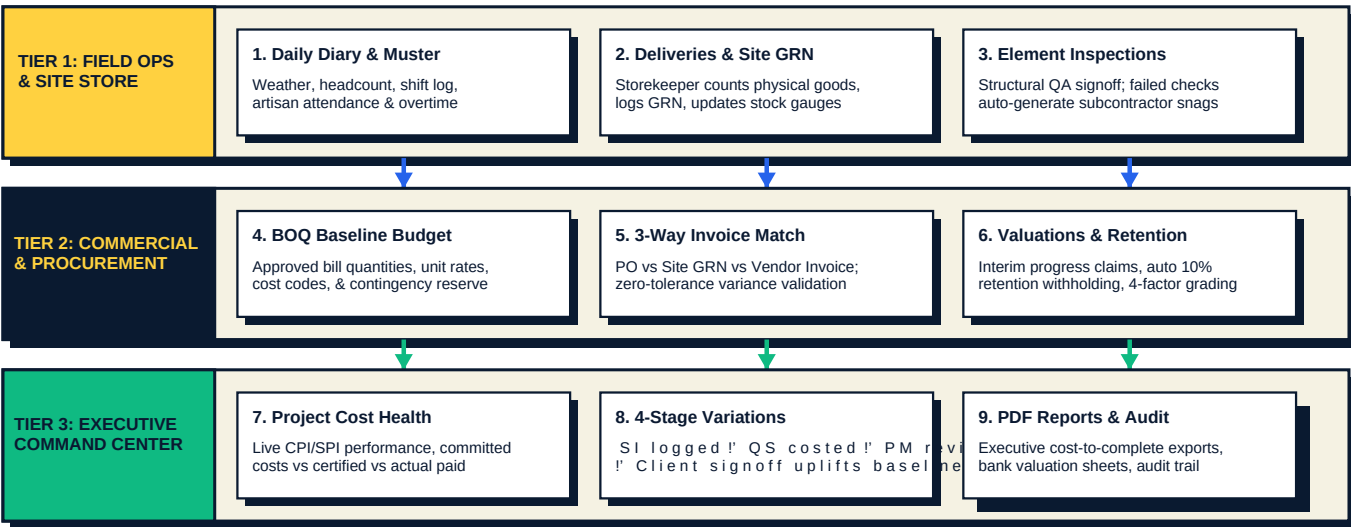
Standard Nigerian building contract clause (JCT/FIDIC) enforced automatically. 10% is withheld from every subcontractor interim certificate, maintaining a secure defect guarantee fund until Practical Completion and final snag sign-off.

TENET 4: DUAL-SIGNOFF GOVERNANCE

Separation of duties protects project funds. Site supervisors record physical facts (deliveries, attendance, snags), while Quantity Surveyors and Project Managers certify valuations and approve change orders before finance issues disbursements.

VISUAL PROCESS GUIDE: END-TO-END SYSTEM DATA FLOW IN COSTVIEW 360

How operational field facts translate into commercial certification and executive financial intelligence:



COMPREHENSIVE EXPLANATION OF THE 10 COSTVIEW 360 CORE SUBSYSTEMS

CostView 360 provides 10 purpose-built modules. Each is designed for specific project personnel to ensure complete commercial control.

Module Name & Path	Primary Users	What To Do: Step-by-Step Actions	System Governance & Value Delivered
1. Commercial Command Center Path: / (Dashboard)	Project Director, Commercial Manager, Contract Administrator	1. Review Top KPIs: Total Budget, Committed Cost, Paid to Date, and Contingency Burn. 2. Monitor Project Health Badge (On Track, At Risk, Over Budget). 3. Check CPI (Cost Performance Index) and SPI (Schedule Performance Index). 4. Review recent site alerts and cash velocity chart.	Delivers instant executive visibility. Detects cost code overruns before funds are committed, giving directors early warning on budget risk.
2. BOQ & Baseline Budget Path: /budget	Senior Quantity Surveyor, Cost Planner	1. Click 'Import Excel / CSV' to upload approved Bill of Quantities. 2. Assign NRM2/CESMM4 cost codes (Substructure, Concrete, MEP, Finishes). 3. Set project contingency reserve (e.g. 5.0%). 4. Review unit rates and total values; click 'Lock Baseline' to establish master control.	Prevents unapproved budget tampering. Once locked, bill items cannot be altered without a formal Variation Order or Revision Notice.
3. Procurement & POs Path: /procurement	Procurement Lead, Quantity Surveyor	1. Review purchase requisitions raised by site supervisors. 2. Obtain and compare 3 competitive supplier RFQ quotes. 3. Issue formal Purchase Order (PO) with binding unit rates and delivery schedules. 4. Route PO for commercial manager financial authorization.	Eliminates rogue purchasing and off-contract pricing. Enforces supplier accountability and pre-approves commercial commitments.
4. Three-Way Invoice Match Path: /procurement (Match Tab)	Financial Controller, Accounts Payable Officer	1. When vendor invoice arrives, open 3-Way Match screen. 2. System links Purchase Order, Site Goods Received Note (GRN), and Invoice. 3. Inspect automated variance calculations (unit price and quantity). 4. If 100% matched, click 'Approve for Payment'. If discrepancy, click 'Raise Dispute'.	Guarantees that finance only pays for items physically verified on site at agreed contract rates. Eliminates phantom invoicing and double-billing.
5. Materials & Stock Control Path: /materials	Site Storekeeper, Materials Engineer	1. Log incoming material trucks; generate Goods Received Note (GRN). 2. Record physical counts (e.g. 500 bags Dangote 42.5R Cement, 12 tonnes rebar). 3. Record daily material issues to artisan crews by BOQ zone/grid. 4. Monitor live stock gauge meters; place reorder requests when thresholds hit yellow.	Reconciles theoretical BOQ quantities against actual physical usage. Prevents site shrinkage, theft, and jobsite material stockouts.
6. Labour Muster & Attendance Path: /labour	Site Supervisor, HR Officer, Site Agent	1. Conduct 07:30 AM morning muster roll. 2. Check in skilled artisans (masons, carpenters, iron benders, plumbers) and daily helpers. 3. Log overtime hours with supervisory sign-off. 4. Reconcile weekly muster records against subcontractor workforce claims.	Eliminates ghost worker payroll fraud. Ensures daily labor expenditure matches certified physical site progress.
7. Site Operations & Diary Path: /site-ops	Resident Engineer, QA/QC Inspector	1. Log daily shift records: weather conditions, delays, and critical work progress. 2. Upload timestamped milestone progress photographs. 3. Conduct element inspections (rebar fixing, formwork, MEP conduit embedment). 4. If inspection fails, system auto-generates a Snag/NCR assigned to the trade contractor.	Provides legally defensible daily site records. Cloud synchronization ensures daily diaries cannot be lost or retrospectively altered.
8. Subcontractor Packages Path: /subcontractors	Quantity Surveyor, Project Manager	1. Award trade packages with contracted scopes and values. 2. When contractor submits monthly valuation, verify progress on site. 3. Enter certified sum; system automatically deducts 10% statutory retention. 4. Generate Interim Payment Certificate and record 4-factor performance grade.	Enforces the 10% statutory retention fund on every claim. Protects the client against subcontractor default and non-performing trades.
9. Variations & Instructions Path: /site-ops (Tabs 7.1 & 7.2)	Architect, Resident Eng, Quantity Surveyor	1. Consultant issues formal Site Instruction (SI) with sketch/drawings. 2. QS links SI to a Variation Order (VO) and computes cost/time delta. 3. Route through 4-stage approval: Valuation! PM Review! Approved. 4. Upon final sign-off, system automatically adjusts master budget baseline.	Eliminates verbal instruction disputes. Ensures all change orders are valued, approved, and tracked before contractor carries out extra work.
10. Reports Studio & Exports Path: /reports	Commercial Director, Client Representative	1. Select project and required reporting period. 2. Choose report type: Executive Cost Summary, Subcontractor Ledger, or Audit Log. 3. Generate high-resolution, branded PDF documents	Transforms complex project figures into boardroom-ready visual reports with transparent cost-to-complete forecasts and audit backing.

Module Name & Path	Primary Users	What To Do: Step-by-Step Actions	System Governance & Value Delivered
		with one click. 4. Distribute to banks, joint-venture partners, and executive board members.	

STANDARD OPERATING PROCEDURES (SOP): SETUP, FIELD ROUTINES & QUALITY

WORKFLOW 1: INITIAL PROJECT SETUP & BOQ BASELINE LOCKING (DAY 1)

- Step 1: Access the Platform — Log in with your Quantity Surveyor (QS) or Commercial Manager account.
- Step 2: Navigate to BOQ & Baseline — Open the 'BOQ & Budget' screen from the main sidebar navigation.
- Step 3: Import Bill of Quantities — Click 'Import Excel / CSV BOQ'. Select your standard bill spreadsheet (supports NRM2, CESMM4, or custom contractor formats with item description, unit, quantity, and contract rate in Naira).
- Step 4: Verify Cost Code Mapping — Confirm items are categorized under major cost codes (e.g. SUB Substructure, CON Concrete Frame, BLK Masonry & Blockwork, FIN Finishes, MEP Mechanical & Electrical, EXT External Works).
- Step 5: Set Contingency Reserve — Establish the project contingency fund (standard 5.0% recommended for price escalation).
- Step 6: Lock Commercial Baseline — Click 'Lock Baseline'. Once locked, initial contract sums are frozen. Any future changes must flow through formal Variation Orders or Revision Notices, establishing an immutable benchmark for cost tracking.

WORKFLOW 2: DAILY SITE OPERATIONS & INSPECTION ROUTINE (CHRONOLOGICAL)

- 07:30 AM — Morning Labour Muster: Site Supervisor opens 'Labour Muster'. Call roll for direct artisan crews and subcontractor gangs. Mark attendance and verify any authorized night overtime hours.
- 09:30 AM — Daily Diary Shift Initiation: Site Engineer opens 'Site Diary'. The system auto-stamps current local weather. Record scheduled activities (e.g., 'Casting second-floor suspended slab columns with 40m³ ReadyMix').
- 12:00 PM — Material Deliveries & Store GRN: Storekeeper inspects arriving trucks at site gate. Verify waybills, count bags, rebar bundles, or aggregates. Log Goods Received Note (GRN) in 'Materials & Stock' against the active PO.
- 03:30 PM — Element QA/QC Inspections: Resident Engineer conducts quality checklists (e.g., rebar spacing, cover blocks, formwork plumbness). If any check fails, the system automatically logs a defect Snag assigned to the trade.
- 05:30 PM — Shift Closeout & Sync: Complete daily shift report, attach milestone progress photos, record any weather delay hours (vital for contractual extension of time claims), and submit for Resident Engineer verification.

VISUAL PROCESS FLOW: DAILY FIELD SITE ROUTINE & QUALITY SNAG MANAGEMENT

Visual execution sequence from morning workforce muster through delivery receipts, quality checks, and defect tracking:

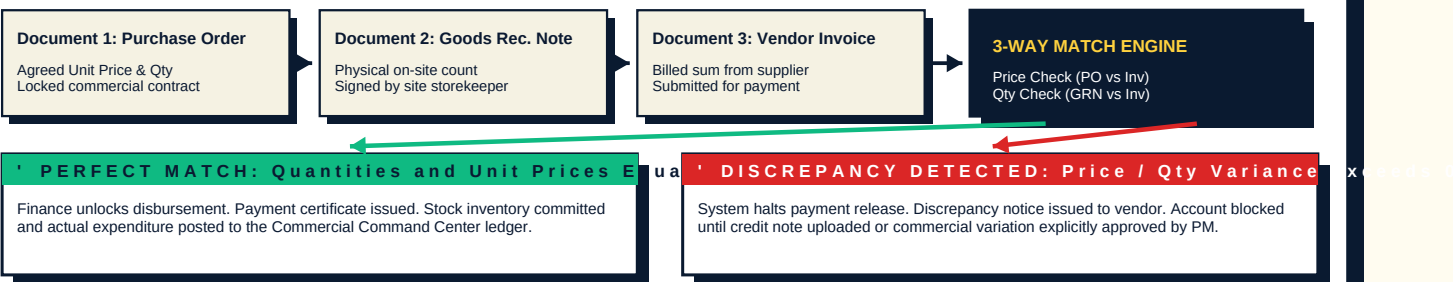


COMMERCIAL WORKFLOWS: 3-WAY MATCHING & 10% RETENTION VALUATION

WORKFLOW 3: PROCUREMENT & 3-WAY INVOICE MATCHING (ZERO OVERPAYMENT)

- Step 1: Requisition Creation — Site supervisor raises material request referencing the exact BOQ line item.
- Step 2: RFQ Solicitation & PO Issuance — Procurement compares 3 supplier quotes. Commercial manager authorizes the Purchase Order (PO) locking unit price (e.g., ₹8,500/bag) and quantity (e.g., 500 bags).
- Step 3: Physical Delivery & Site GRN — When goods arrive on site, the Storekeeper inspects the physical shipment, counts bags, and issues a Goods Received Note (GRN) (e.g., actual verified count: 480 bags).
- Step 4: Invoice Entry & 3-Way Match — Vendor submits invoice (e.g., billing 500 bags at ₹8,800/bag). The system automatically compares: PO (500 @ ₹8,500) vs GRN (480) vs Invoice (500 @ ₹8,800). The system flags a quantity delta (20 bags short) and a price delta (₹300/bag).
- Step 5: Automated Variance Resolution — The system detects a quantity delta (20 bags short) and a price delta (₹300/bag). CostView 360 immediately locks the invoice, blocking payment release until the vendor issues an amended credit note.

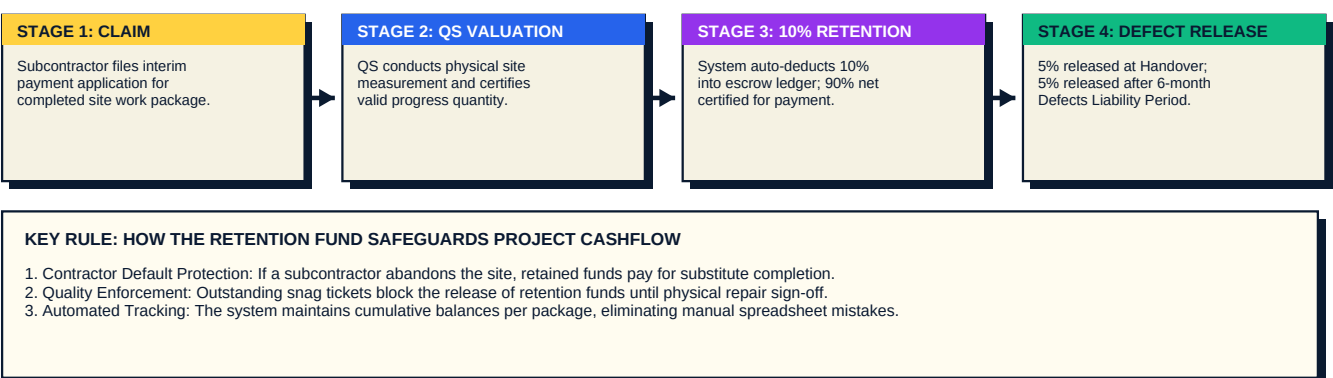
VISUAL DATA FLOW: AUTOMATED 3-WAY INVOICE RECONCILIATION GATE



WORKFLOW 4: SUBCONTRACTOR INTERIM VALUATIONS & 10% RETENTION LIFECYCLE

- Step 1: Subcontractor Milestone Claim — Trade contractor submits monthly application (e.g. Zenith Piling: ₹1,85,00,000) for interim payment.
- Step 2: Physical Site Measurement — Quantity Surveyor visits site, verifies borehole depths and piles cast against drawings.
- Step 3: Certification & Automatic 10% Deduction — QS enters certified sum in 'Subcontractor Packages' and clicks 'Certify & Apply 10%'. CostView 360 immediately isolates ₹1,85,00,000 into the cumulative retention escrow ledger and approves the net ₹16,65,00,000 for financial payment.
- Step 4: Four-Factor Contractor Grading — QS submits monthly performance rating across 4 factors: Schedule Punctuality, Quality Compliance, Site Safety, and Commercial Cooperation. Maintains an objective historical scorecard.
- Step 5: Handover & Retention Release — At Practical Completion, 5% (first half) is released upon snag list clearance. The remaining 5% is released following the 6-month Defects Liability Period signoff by the Resident Engineer.

VISUAL PROCESS MAP: 10% RETENTION DEDUCTION & DEFECT RELEASE CYCLE



CHANGE MANAGEMENT GOVERNANCE & FIELD TROUBLESHOOTING MATRIX

WORKFLOW 5: 4-STAGE VARIATION ORDER APPROVAL PIPELINE

Stage 1: Site Instruction (SI) Issued — Architect or Resident Engineer logs formal SI in '7.1 Site Instructions'.

Stage 2: QS Cost & Delay Valuation — Quantity Surveyor estimates price delta (!) and time extension impact.

Stage 3: Project Manager Review — PM evaluates commercial impact against the 5.0% contingency reserve fund.

Stage 4: Formal Client Approval & Baseline Uplift — Client gives formal approval. CostView 360 immediately updates the master budget baseline, commits funds, and authorizes the site team to execute the additional work package.

STANDARD IN-APP STATUS BADGES & MEANINGS

Badge Name	Visual Color	Meaning & Trigger Condition	Required User Action
ON TRACK	Green (#10B981)	Expenditure is within approved BOQ baseline and contingency buffer.	Continue standard execution; no commercial intervention needed.
AT RISK	Amber (#D97706)	Cost code has exceeded 85% of budget or variance delta exceeds 5%.	QS must review upcoming commitments and inspect material wastage.
OVER BUDGET	Red (#DC2626)	Committed costs exceed 100% of approved baseline allocation.	Requires formal Variation Order or contingency draw-down approval.
MATCHED	Green (#10B981)	PO unit rate, GRN physical count, and vendor invoice agree 100%.	Finance authorizes invoice for payment disbursement.
DISCREPANCY	Red (#DC2626)	Unit price is higher than PO, or billed quantity exceeds physical GRN.	System halts payment. Procurement contacts vendor for credit note.
RETAINED (10%)	Purple (#9333EA)	Statutory 10% withheld from interim certificate into escrow ledger.	Held securely until Practical Completion and defect snag clearance.
SNAG OPEN	Amber (#D97706)	Defect ticket logged from failed site inspection or supervisor check.	Contractor must rectify defect; blocks final milestone certification.
SNAG CLOSED	Green	Resident Engineer re-inspected physical repair and verified	Rectification recorded in diary; releases trade milestone

FIELD TROUBLESHOOTING MATRIX: WHAT TO DO IN REAL SITE SITUATIONS

Operational Site Scenario	Immediate In-App Action Required	Commercial Protection Delivered
Supplier delivers partial order (e.g., 80 bags delivered vs 100 on PO)	Storekeeper enters exact physical count (80) on the Goods Received Note (GRN). Do not enter 100.	3-Way Match locks invoice if vendor bills for 100. Eliminates paying for 20 undelivered bags.
Vendor invoice unit price is higher than agreed Purchase Order rate	Open 3-Way Match, review red price variance flag, and click 'Raise Dispute' with note.	Halts payment disbursement until supplier reissues invoice with contracted rate.
Subcontractor objects to 10% statutory retention deduction	Refer contractor to contract retention terms. Show cumulative retention escrow ledger balance.	Maintains mandatory client defect guarantee fund per Nigerian construction industry standards.
Torrential rain halts concrete pour or earthworks on site	Open 'Site Diary', record exact rain duration and delay hours, and attach site photos.	Provides contemporaneous legal documentation required to evaluate Extension of Time (EOT) claims.
Material stock level indicator turns yellow or red	Open 'Materials & Stock' and click 'Create Reorder Request' before stock runs out.	Prevents costly site shutdowns and emergency spot-buying at inflated retail prices.
Structural inspection fails (e.g., honeycombing on column)	Resident Engineer marks inspection 'Failed' and uploads photo. System auto-generates Snag ticket.	Blocks trade milestone payment certification until contractor performs approved structural repair.